

Cytiva Xcellerex XDR-500 - 100-500 L working volume

cytiva-xdr-500

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	100 L	Source-supported
Maximum working volume	500 L	Source-supported
Nominal vessel / bag size	500 L	Source-supported
Total vessel volume loaded into the form	675.9 L	Derived / estimated
Initial working volume loaded into the form	100 L	Derived / estimated
Product-contact material	Ultra-low-density polyethylene (ULDPE) contact layer	Source-supported
Height : diameter	2	App assumption
Internal / chamber diameter	0.6828 m	Derived / estimated
Liquid height at maximum working volume	1.37 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	Source-supported
Impeller count	1	Source-supported
Impeller diameter	0.2253 m	Derived / estimated
Power number	0.4	App assumption
Initial agitation	80 rpm	App assumption
Maximum agitation	200 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 100 L

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 6; Table 1, XDR-500

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 100 L.

Limitations: Model-specific working range; XDR-50 minimum is 25 L, not a family 20% assumption.

Local original: [Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 6](#)

[Manufacturer publication](#)

Maximum working volume - 500 L

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 6; Table 1, XDR-500

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: Model-specific working range; XDR-50 minimum is 25 L, not a family 20% assumption.

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Nominal vessel / bag size - 500 L

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 6; Table 1, XDR-500

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: Model-specific working range; XDR-50 minimum is 25 L, not a family 20% assumption.

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Total vessel volume loaded into the form - 675.9 L

Derived / estimated | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 6; Table 1, working volume and headspace

Rule: total = maximum working volume + published headspace

Selection / calculation: $500 + 175.9 \text{ L}$

Limitations: Total calculated by adding the published headspace to maximum working volume.

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[Manufacturer publication](#)

Initial working volume loaded into the form - 100 L

Derived / estimated | Form-population rule

Rule: initial working volume = $\min(\text{minimum working volume}, \text{maximum working volume})$

Selection / calculation: $\min(100 \text{ L}, 500 \text{ L}) = 100 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Ultra-low-density polyethylene (ULDPE) contact layer

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 3; Disposable bag and impeller description

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Ultra-low-density polyethylene (ULDPE) contact layer.

Limitations: Single bottom-mounted pitched impeller. The stainless-steel support vessel is not the product-contact layer.

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[Manufacturer publication](#)

Height : diameter - 2

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].heightDiameterRatio = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.6828 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V) / (\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.5 \text{ m}^3) / (\pi \times 2)} = 0.6827841 \text{ m}$; the form displays 0.6828 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 1.37 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.6827841 \text{ m} \times 2 = 1.365568 \text{ m}$; the form displays 1.37 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Pitched-blade turbine

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 3; Disposable bag and impeller description

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: Single bottom-mounted pitched impeller. The stainless-steel support vessel is not the product-contact layer.

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[Manufacturer publication](#)

Impeller count - 1

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 3; Disposable bag and impeller description

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.

Limitations: Single bottom-mounted pitched impeller. The stainless-steel support vessel is not the product-contact layer.

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Impeller diameter - 0.2253 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.6827841 \text{ m}, 0.001 \text{ m}) = 0.2253187 \text{ m}$; the form displays 0.2253 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 0.4

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].powerNumber = 0.4

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The power number is an empirical generic value used in $P = N_p N_3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 80 rpm

App assumption | Large mammalian-vessel scaling rule

Rule: $\text{baseRpm} = \text{maxVolume} \geq 1000 \text{ L} ? 55 : 80$

Selection / calculation: 500 L is not at least 1000 L -> 80 rpm

Limitations: The simulator lowers its generic starting agitation at larger scale. This is an internal scale heuristic, not a manufacturer limit. This is the initial agitation setting loaded into the form.

Maximum agitation - 200 rpm

App assumption | Large mammalian-vessel scaling rule

Rule: $\text{maxRpm} = \text{maxVolume} \geq 1000 \text{ L} ? 150 : 200$

Selection / calculation: 500 L is not at least 1000 L -> 200 rpm

Limitations: The simulator lowers its generic agitation ceiling at larger scale. This is an internal scale heuristic, not a manufacturer limit. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{baseVvm} = 0.05$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{maxVvm} = 0.3$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{baffled} = \text{false}$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{maxPressure} = 0.5$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: $\text{PROFILE_DEFAULTS}["\text{mammalian}"].\text{operatingPressure} = 0.05$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

Large XDR presets use a generic mammalian stirred-tank profile unless a model-specific field is stored.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Cytiva Xcellerex XDR 50 to 2000 L data file

cytiva-xcellerex-xdr-50-to-2000-l-data-file-d7b9c0ca50bd.pdf

SHA-256: d7b9c0ca50bd6d6eff70df4dbd3d693d2c0a710badff26f8dece1f2333125f96